

module2

day 8

Absolutely 🙌 **ACL is another big final-exam topic.** Your material is very detailed, so I'll reduce it to the **important concepts, numbers, commands, rules, and 2 easy + 2 hard examples.**

📦 ACL FINAL EXAM – HIGH-YIELD NOTES

1. ☆ What is an ACL?

ACL = Access Control List

An ACL is a list of **permit and deny rules** used to control network traffic.

Simple:



ACL = Who is allowed to go where?

Example:

PC1 → Router → Server



ACL



Permit / Deny

Copy

ACL can control traffic based on things such as:

- Source IP
- Destination IP
- Protocol
- Port number



2. ☆ The Most Important ACL Rule

ACLs are processed:



TOP → BOTTOM

And:



FIRST MATCH WINS

Example:

```
1. deny host 192.168.1.10
2. permit 192.168.1.0 0.0.0.255
```

 Copy

If traffic comes from:

```
192.168.1.10
```

[Copy](#)

Rule #1 matches.

× Denied.

The router **does not continue** to rule #2.

Memorize:



First match = final decision.



3. ☆ Implicit Deny

Every ACL has an invisible rule at the end:

```
deny any
```

[Copy](#)

or conceptually:

```
deny ip any any
```

[Copy](#)

Therefore:



If a packet doesn't match a permit rule, it is **DENIED**.

Example:

```
1. permit host 192.168.1.10  
2. permit host 192.168.1.20  
3. implicit deny
```

Copy

Traffic from:

192.168.1.30

Copy

× Denied.

Exam trap

If you create:

```
access-list 10 deny host 192.168.1.10
```

Copy

you have effectively denied **everything else too**, because of the implicit deny.

4. ☆ Standard vs Extended ACL

This is probably the **most important ACL comparison**.

Feature	Standard ACL	Extended ACL
Checks	Source IP only	Source + Destination + Protocol + Port
Number	1–99, 1300–1999	100–199, 2000–2699
Placement	Near destination	Near source
Example	Block a host	Block HTTP from one network
Detail	Simple	More specific

Easy memory



Standard = Source



Extended = Everything



5. ☆ Why Placement Matters

Standard ACL

Place **close to the destination**.

Why?

Because standard ACL only knows the **source**.

If you put it too close to the source, you might accidentally block that source from reaching other destinations.



Extended ACL

Place **close to the source**.

Why?

Because it can specifically identify:

- Source
- Destination
- Protocol
- Port

So unwanted traffic can be stopped early.

Memorize:

Standard → Destination



Extended → Source



6. ☆ Standard ACL Numbers

Standard numbered ACL:

1–99
1300–1999

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Example:

```
access-list 10 deny host 192.168.1.10
```

Copy



7. ☆ Extended ACL Numbers

Extended numbered ACL:

100–199
2000–2699

Copy

Example:

```
access-list 110 deny tcp any host 172.16.30.2 eq 80
```

Copy

Meaning:



Deny TCP traffic from anywhere to 172.16.30.2 on port 80 .



8. ☆ Named ACL

Instead of using a number, you can use a name.

Example:

```
ip access-list extended WEB_FILTER
```

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Then:

```
deny tcp any host 192.168.1.10 eq 80
permit ip any any
```

[Copy](#)

Advantage

Named ACLs are easier to:

- Read
- Edit
- Insert rules
- Delete individual rules



9. ☆ `host` **and** `any`

These are very important shortcuts.

`host`

```
host 192.168.1.10
```

[Copy](#)

means:

```
192.168.1.10 0.0.0.0
```

[Copy](#)

It means **one specific IP address**.



any

any

Copy

means:

0.0.0.0 255.255.255.255

Copy

It means:



Any IP address

Remember:



host = one



any = everyone



10. ☆ ACL Direction

ACLs can be applied:

IN

Traffic enters the interface.

```
ip access-group 10 in
```

 Copy

OUT

Traffic leaves the interface.

```
ip access-group 10 out
```

 Copy

Easy picture:

IN
↓
—— [ROUTER] ——
↓
OUT

Copy



11. ☆ One ACL Per Direction

For one interface, you can have:

1 inbound ACL
+
1 outbound ACL

Copy

So maximum:



2 IP ACLs per interface

Example:

G0/0
|— ACL 10 IN

Copy

You cannot have two separate inbound IP ACLs on the same interface.



12. ☆ ip access-group

Used to apply an ACL to an interface.

Example:

```
interface g0/0
ip access-group 10 in
```

Copy

Meaning:



Apply ACL 10 to traffic entering G0/0.



13. ☆ ACLs Don't Filter Router- Originated Traffic

ACLs applied to an interface filter traffic **passing through the router**.

They don't normally filter traffic generated directly by the router itself.

Example:

```
PC → Router → Server
```

Copy

ACL can filter this transit traffic.

But:

```
Router → Server
```

Copy

traffic generated by the router itself is not treated the same way as transit traffic.



14. ☆ Standard ACL Configuration

Deny one host

```
access-list 10 deny host 192.168.1.10
```

Copy

Permit a subnet

```
access-list 10 permit 192.168.1.0 0.0.0.255
```

[Copy](#)

Permit everything

```
access-list 10 permit any
```

[Copy](#)

Apply

```
interface g0/0  
ip access-group 10 in
```

[Copy](#)

15. ☆ Extended ACL Configuration

General syntax:

```
access-list 100 permit|deny protocol source destination
```

[Copy](#)

For ports:

```
access-list 100 deny tcp source destination eq port
```

 Copy

Example:

```
access-list 110 deny tcp any host 192.168.1.10 eq 80
```

 Copy

Means:



Block TCP HTTP traffic from anywhere to 192.168.1.10 .



16. ☆ Common Ports

These are worth memorizing for the exam.

Service	Protocol	Port
FTP Data	TCP	20
FTP Control	TCP	21
SSH	TCP	22
Telnet	TCP	23
SMTP	TCP	25

Service	Protocol	Port
DNS	UDP	53
HTTP	TCP	80
SNMP	UDP	161

Most important:

FTP = 20/21
 SSH = 22
 Telnet = 23
 DNS = 53
 HTTP = 80

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17. ☆ Port Operators

Extended ACLs can use:

Operator	Meaning
eq	Equal to
gt	Greater than
lt	Less than
neq	Not equal

Operator	Meaning
range	Range of ports

Example:

```
eq 80
```

 Copy

means:



Exactly port 80.

Example:

```
range 20 21
```

 Copy

means:



Ports 20 through 21.



18. ☆ established

Example:

```
access-list 102 permit tcp any any established
```



This is commonly used to permit TCP traffic associated with an **already-established TCP connection**.

It is often used to allow return TCP traffic while restricting unsolicited inbound connections.

Exam idea:



established = allow returning TCP traffic

⚠ It is not a full modern stateful firewall. It relies on TCP header flags.



19. ☆ VTY Security

VTY lines control remote management such as:

- SSH
- Telnet

Use:

```
access-class
```



NOT:

```
ip access-group
```

Copy

Example:

```
line vty 0 15  
access-class 10 in
```

Copy

Very important:



Interface ACL → ip access-group



VTY ACL → access-class



20. ☆ ACL Verification Commands

Show ACLs

```
show access-lists
```

[Copy](#)

Shows ACL rules and match counters.



Show IP ACL

```
show ip access-lists
```

[Copy](#)

or:

```
show ip access-lists 10
```

[Copy](#)

See which ACL is applied to an interface

```
show ip interface
```

[Copy](#)

See complete configuration

```
show running-config
```

[Copy](#)

Reset counters

```
clear access-list counters
```

[Copy](#)

21. ☆ ACL Hit Counters

ACLs count how many packets matched each rule.

Example:

```
10 permit host 192.168.1.10  
(25 matches)
```

[Copy](#)

This tells you that 25 packets matched that statement.

Useful for troubleshooting.



22. ☆ Explicit Deny

You can add:

```
deny ip any any
```

[Copy](#)

at the bottom.

Technically, the ACL already has an **implicit deny**, but an explicit deny makes the policy visible and allows you to see its match counter.



23. ☆ ACL Rule Ordering

Always put **specific rules first**.

Bad:

```
permit ip any any  
deny host 192.168.1.10
```

[Copy](#)

The deny will **never be reached**.

Why?

Because:

```
permit ip any any
```

 Copy

matches everything first.

Correct:

```
deny host 192.168.1.10  
permit ip any any
```

 Copy

Golden rule:



Specific → General



● EASY EXAMPLE 1 – Block One PC

Question

PC1:

192.168.1.10

 Copy

You want to block PC1 from passing through the router.

Configuration

```
access-list 10 deny host 192.168.1.10
access-list 10 permit any
```

 Copy

Apply:

```
interface g0/0
ip access-group 10 in
```

 Copy

What happens?

```
PC1 → Router
192.168.1.10
↓
ACL 10
↓
DENY ✕
```

 Copy

Other hosts:

```
192.168.1.20 → permit []  
192.168.1.30 → permit []
```

[Copy](#)

Because of:

```
permit any
```

[Copy](#)

● EASY EXAMPLE 2 – Block HTTP

Question

Web server:

```
192.168.10.10
```

[Copy](#)

Block HTTP access to it from anywhere.

HTTP:

```
TCP 80
```

[Copy](#)

ACL:

```
access-list 110 deny tcp any host 192.168.10.10 eq 80
access-list 110 permit ip any any
```

[Copy](#)

Apply:

```
interface g0/0
ip access-group 110 in
```

[Copy](#)

Result

```
Internet
|
| TCP/80
↓
192.168.10.10
× DENIED
```

[Copy](#)

Other traffic:

```
SSH → allowed
DNS → allowed
ICMP → allowed
```

[Copy](#)

because of:

```
permit ip any any
```

Copy



HARD EXAMPLE 1 – Rule Ordering

Suppose you have:

```
access-list 110 permit ip any any  
access-list 110 deny tcp host 192.168.1.10 host 192.168.2.10 eq 80
```

Copy

Question:

Will HTTP from:

```
192.168.1.10
```

Copy

to:

```
192.168.2.10
```

Copy

be blocked?

Answer:

× **NO**

Why?

ACL processes top-to-bottom.

First rule:

```
permit ip any any
```

 Copy

matches.

So the router immediately permits the packet.

The second rule is never checked.

Correct ACL:

```
access-list 110 deny tcp host 192.168.1.10 host 192.168.2.10 eq 80  
access-list 110 permit ip any any
```

 Copy

Now:

```
Specific deny  
↓
```

 Copy

General permit

☐ Correct.



HARD EXAMPLE 2 – Design an ACL

Situation

You have:

LAN: 192.168.1.0/24
Web Server: 192.168.2.10

 Copy

Requirement:

1. Block LAN users from HTTP to the Web Server.
2. Allow all other traffic.
3. Use an extended ACL.
4. Apply it close to the source.

Step 1 – Extended ACL

Because we need:

- Source

- Destination
- Protocol
- Port

Use extended ACL.

Step 2 – HTTP

HTTP:

TCP 80

 Copy

Step 3 – Configuration

```
access-list 110 deny tcp 192.168.1.0 0.0.0.255 host 192.168.2.10 eq 80
access-list 110 permit ip any any
```

 Copy

Step 4 – Apply close to source

```
interface g0/0
ip access-group 110 in
```

 Copy

Result:

Copy

192.168.1.0/24

|

| HTTP

↓

ACL 110

|

X

↓

192.168.2.10

HTTP:

× Blocked

Other traffic:

□ Allowed



☆ Anti-Spoofing

On an Internet-facing interface, you can block packets pretending to come from invalid/private source networks.

Common examples:

Private 10/8

Copy

10.0.0.0/8

Wildcard:

0.255.255.255

 Copy

Private 172.16/12

172.16.0.0/12

 Copy

Wildcard:

0.15.255.255

 Copy

Private 192.168/16

192.168.0.0/16

 Copy

Wildcard:

0.0.255.255

 Copy

Loopback

127.0.0.0/8

Copy

Wildcard:

0.255.255.255

Copy

Multicast

224.0.0.0/4

Copy

Wildcard:

15.255.255.255

Copy



☆ Anti-Spoofing ACL Example

Applied inbound on an Internet interface:



```
access-list 121 deny ip 10.0.0.0 0.255.255.255 any
access-list 121 deny ip 172.16.0.0 0.15.255.255 any
access-list 121 deny ip 192.168.0.0 0.0.255.255 any
access-list 121 deny ip 127.0.0.0 0.255.255.255 any
access-list 121 deny ip 224.0.0.0 15.255.255.255 any
access-list 121 permit ip any any
```

Apply:



```
interface s0/0/0
ip access-group 121 in
```

Why?

A packet arriving from the Internet with:



Source = 192.168.1.10

is suspicious because private addresses should not normally appear as legitimate Internet source addresses.

× Drop it.



☆ **ACL Logging**

You can use:

```
deny ip any any log
```

Copy

This records matching traffic.

⚠ Important

Too much logging can consume router CPU.

So:



Use logging carefully.



DMZ

DMZ = Demilitarized Zone

A DMZ is a network zone commonly used for public-facing servers.

Example:

Internet



Router

Copy



The DMZ separates public services from the trusted internal network.

Example:

Internet → Web Server

Copy

might be allowed.

But:

Internet → Internal PCs

Copy

should normally be restricted.



☆ Inter-VLAN ACLs

If you're doing inter-VLAN routing with router subinterfaces, ACLs can be applied directly to the subinterfaces.

Example:

```
interface g0/0.10
ip access-group 110 in
```

[Copy](#)

This allows you to control traffic entering VLAN 10's routed interface.



ACL FINAL EXAM CHEAT SHEET

Memorize this block:

ACL



Access Control List

Processing



Top → Bottom

First Match Wins

Implicit Deny at End

STANDARD



Source IP ONLY

Numbers: 1–99, 1300–1999

Place near DESTINATION

EXTENDED

[Copy](#)

Source + Destination + Protocol + Port

Numbers: 100–199, 2000–2699

Place near SOURCE

host

↓

One IP

any

↓

Any IP

IN

↓

Entering interface

OUT

↓

Leaving interface

Interface ACL

↓

ip access-group

VTY ACL

↓

access-class

Maximum

↓

1 IN + 1 OUT per interface

HTTP

↓

TCP 80

HTTPS

↓

TCP 443

FTP

↓

TCP 20/21

SSH

↓

TCP 22

Telnet

↓

TCP 23

DNS

↓

UDP 53

SMTP

↓

TCP 25

SNMP

↓

UDP 161

Verification

↓

show access-lists

show ip access-lists

```
show ip interface
show running-config
clear access-list counters
```

The 10 ACL Facts I'd Memorize First

1. ACL = permit/deny traffic
2. Top-to-bottom
3. First match wins
4. Implicit deny at the end
5. Standard = source IP only
6. Extended = source + destination + protocol + port
7. Standard → near destination
8. Extended → near source
9. Interface = `ip access-group`
10. VTY = `access-class`

One-line memory trick



Standard = Source → Destination

Extended = Everything → Source

First match wins → Implicit deny → Permit what you need.

That little chain will save you a *lot* of exam pain. 😊